

Fig. 5.5. *Variation of density of electronic states with energy for a one-dimensional metallic crystal. At* 0 K, *all the states upto the Fermi level are filled.*

The plot of $D(E)$ versus E is shown in Fig. 5.5. It indicates that all the levels present below the Fermi level are filled and all those present above it are empty. This type of situation, in fact, exists at absolute zero. Thus at 0 K, the Fermi level divides the filled and unfilled levels in the metallic crystals.

5.2.2 Free Electron Gas in Three Dimensions

Consider now the behaviour of the free electron gas in a three-dimensional metallic crystal. The potential of the electrons inside the crystal is constant and may be taken as zero, whereas it has a large value outside the crystal. For simplicity, the three-dimensional crystal may be regarded as a cubical box having length of the edge equal to L. The free particle Schrodinger equation in three dimensions is

$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}\right)\psi_k(\mathbf{r}) = E_k\psi_k(\mathbf{r})$$

or
$$\nabla^2\psi_k(\mathbf{r}) + \frac{2m}{\hbar^2}E_k\psi_k(\mathbf{r}) = 0 \qquad (5.19)$$

where
$$\nabla^2 \equiv \frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}$$

is the Laplacian operator and E_k is the total energy (kinetic energy in the present case) of the electron in the k-state. Since the electrons are confined to a cubical box of edge L, the solution to Eq. (5.19) is just an extension of the one-dimensional normalized wave function (5.11), i.e.,

$$\psi_k(\mathbf{r}) = \sqrt{\frac{8}{L^3}}\sin\left(\frac{\pi n_x x}{L}\right)\sin\left(\frac{\pi n_y y}{L}\right)\sin\left(\frac{\pi n_z z}{L}\right) \qquad (5.20)$$

where n_x, n_y and n_z are positive integers and $\sqrt{8/L^3}$ is the normalising constant. This equation represents a standing wave solution. It is, however, more convenient to work with the plane travelling wave solution of the type

$$\psi_k(\mathbf{r}) = A\, e^{i\mathbf{k}.\mathbf{r}} = Ae^{i\left(k_x x + k_y y + k_z z\right)} \qquad (5.21)$$

where $$k^2 = \frac{2m}{\hbar^2} E_k \tag{5.22}$$

and A is an arbitrary constant. Such wave functions must satisfy the periodic boundary conditions, i.e., they must be periodic in x, y and z with period equal to L. These boundary conditions are

$$\begin{aligned} \psi(x + L, y, z) &= \psi(x, y, z) \\ \psi(x, y + L, z) &= \psi(x, y, z) \\ \psi(x, y, z + L) &= \psi(x, y, z) \end{aligned} \tag{5.23}$$

An application of the first boundary condition in (5.23) to the wave function (5.21) yields

$$\exp[i\{k_x(x + L) + k_y y + k_z z\}] = \exp\,[i\{k_x x + k_y y + k_z z\}]$$

or

$$\exp(ik_x L) = 1$$

or

$$k_x = 0, \pm \frac{2\pi}{L}, \pm \frac{4\pi}{L}, \text{------} \pm \frac{2n\pi}{L} \tag{5.24}$$

Similar results are obtained for k_y and k_z. This implies that any component of k is of the form $\frac{2n\pi}{L}$, where n is a positive or negative integer. These three components of k also form the quantum numbers of the problem in addition to the quantum number m_s which represents the spin direction. Thus the state of an electron is specified completely by a set of four quantum numbers k_x, k_y, k_z and m_s.

The allowed eigen values of the state or orbital with wave vector **k** are obtained from Eq. (5.22) as

$$E_k = \frac{\hbar^2 k^2}{2m} = \frac{\hbar^2}{2m}\left(k_x^2 + k_y^2 + k_z^2\right) \tag{5.25}$$

where the magnitude of the wave vector **k** is related to the wavelength λ as

$$k = \frac{2\pi}{\lambda} \tag{5.26}$$

Thus it is easy to see that the energy spectrum consists of discrete energy levels. These levels are generally very close to each other (energy difference ~ 10^{-15} eV) and may be regarded as continuous for most of the purposes.

These levels are also said to be *quasi-continuous*. In fact, the spacing between the levels depends upon the dimensions of the box. For dimensions of the order of a centimetre, the levels are so closely spaced that they almost form a continuum as predicted by the classical mechanics. If, however, the box has atomic dimensions, the energy levels are widely spaced and discrete.

The constant A in Eq. (5.21) can be determined by using the normalization condition, i.e.,

$$\int_0^V \psi^*(\mathbf{r})\,\psi(\mathbf{r})\,dV = 1$$

or
$$\int_0^L\int_0^L\int_0^L A^2 e^{-i\mathbf{k}.\mathbf{r}} e^{i\mathbf{k}.\mathbf{r}}\,dx\,dy\,dz = 1$$

or
$$A = \left(\frac{1}{L^3}\right)^{1/2} = \left(\frac{1}{V}\right)^{1/2}$$

Hence the normalized wave function is written as

$$\psi_k(\mathbf{r}) = \left(\frac{1}{V}\right)^{1/2} e^{i\mathbf{k}.\mathbf{r}} \tag{5.27}$$

The distribution of electrons among the allowed energy levels and the density of states are described in the following sections.

(i) Filling of Energy Levels

As described earlier, the distribution of electrons in the allowed energy levels follows the Pauli's exclusion principle. In the present case, we have a set of four quantum numbers k_x, k_y, k_z and m_s which represents a state of the system. Apparently, each state can be occupied by at the most one electron. It also follows from the expression (5.25) that a set of values of k_x, k_y and k_z defines an energy level. Now since for each set of k_x, k_y and k_z the fourth quantum number can have two values $\pm 1/2$, it follows that each energy level contains two quantum states of orbitals and hence can accommodate a maximum of two electrons, one with spin up and the other with spin down. In other words, each energy level is doubly degenerate. Thus a total of N non-interacting electrons at 0 K can be filled in $N/2$ energy levels. The topmost filled level is the $(N/2)$th level and all the levels lying

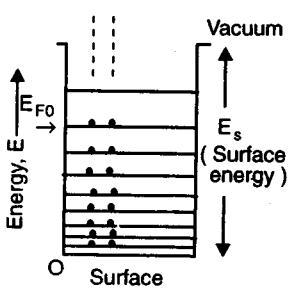


Fig. 5.6. *Sommerfeld's free electron model at* 0 K. *All the levels below* E_{F_0} *are filled and all those above* E_{F_0} *are empty.*

above it are empty. This level is, therefore, the *Fermi level* as it divides the filled and empty levels at 0 K. The energy of the Fermi level, or the Fermi energy, is denoted by E_{F0}. The distribution of electrons among the energy levels present inside the box at 0 K is shown in Fig. 5.6. Thus we note that, unlike the classical theory, the Sommerfeld's quantum theory does not allow the condensation of all the electrons into the state of zero energy even at absolute zero. These electrons are rather distributed among the discrete energy levels having energies ranging from 0 to E_{F0}.

We now discuss the effect of temperature on the occupancy of energy levels. It is apparent that, for temperatures greater than 0 K, the Fermi level may not be the topmost filled level since some of the electrons from the filled energy levels may be excited to the higher levels. Thus some of the levels below E_{F0} would be empty while some above it would be occupied. The probability that a particular quantum state of energy E is occupied at a temperature T is given by the so called *Fermi function*

$$f(E) = \frac{1}{\exp\left(\frac{E - E_F}{k_B T}\right) + 1} \tag{5.28}$$

where E_F is the Fermi energy. The plot of $f(E)$ versus E is shown in Fig. 5.7 for different temperatures.

At absolute zero, the Eq. (5.28) gives

$$\begin{aligned} f(E) &= 1 \quad \text{for} \quad E \le E_{F0} \\ &= 0 \quad \text{for} \quad E > E_{F0} \end{aligned} \tag{5.29}$$

This indicates that all the energy states below E_{F0} are occupied and all the states above it are empty, i.e., the Fermi distribution function is a step function. Thus, at 0 K, the Fermi level E_{F0} represents the highest filled energy level.

For temperatures greater than 0 K but less than the melting point of the metal such that $k_B T << E_F$, the distribution function loses its step character. The probability of occupation, $f(E)$, decreases gradually from 1 to 0 near E_F. This indicates that some of the states below E_F are empty while some others above it are filled. This is because some

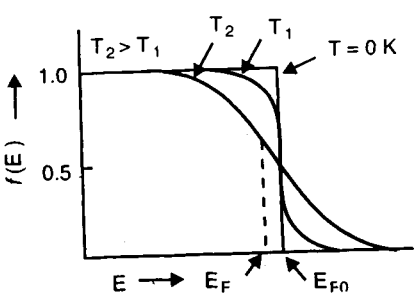


Fig. 5.7. *Schematic representation of the Fermi distribution function for three different temperatures. The variation of the Fermi energy with temperature is also shown.*

of the electrons from the energy states below E_F gain thermal energy and get excited to the states above E_F. At $E = E_F$, Eq. (5.28) gives

$$f(E_F) = \frac{1}{2}$$

Thus, for temperatures greater than 0 K, the Fermi level may be defined as the level where the probability of occupation is 1/2. Unlike E_{Fo}, it is not the topmost filled level; instead, it lies between the filled levels and empty levels. Apparently, the position of the Fermi level is not fixed, but changes with temperature. An approximate relationship between E_F and E_{Fo} is given as follows :

$$E_F \cong E_{Fo}\left[1-\frac{\pi^2}{12}\left(\frac{k_BT}{E_{F_o}}\right)^2\right] \tag{5.30}$$

However, in case of metals, since the spacing between the levels is quite small ($\sim 10^{-19}$ eV), the highest filled energy level is usually taken as the Fermi level.

For energies below E_F such that $(E_F - E) >> k_BT^*$, $f(E) \cong 1$, provided T does not differ much from 0 K. Thus it is only in the vicinity of E_F minus a few k_BT that $f(E)$ becomes less than unity, i.e., some of the states below E_F would be empty and some above E_F would be filled. It, therefore, follows that as the temperature is raised above 0 K, all the electrons would not gain energy as expected classically, but only those which lie within an energy range k_BT below the Fermi level can do so. This is because the electrons present near the Fermi level can jump to the vacant higher energy states after acquiring energy of the order of k_BT, but the electrons present well below the Fermi level cannot do so due to the non-availability of empty states within the range k_BT. Thus, according to quantum mechanics, only a small fraction of the electrons can gain thermal energy and get excited to the energy states. The value of this fraction is $\frac{k_BT}{E_F} \cong 0.01$ at room temperature for $E_F = 3.0$ eV.

(ii) Density of Available Electron States, D(E)

As defined earlier, the density of states, $D(E)$, is the total number of available electronic states (or orbitals) per unit energy range at energy E. To obtain the expression for $D(E)$, we consider the linear momentum **p** which, in quantum mechanics, is represented by the operator

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* According to the classical mechanics, k_BT represents the energy acquired by a particle at a temperature T K. At room temperature, $k_BT \sim 0.025$ eV.

$$\mathbf{p} = -i\hbar\nabla$$

When operated on Eq. (5.21), it gives

$$\mathbf{p}\psi_k(\mathbf{r}) = -i\hbar\nabla\psi_k(\mathbf{r}) = \hbar\mathbf{k}\psi_k(\mathbf{r}) \tag{5.31}$$

This indicates that the plane wave function ψ_k is an eigen function of the linear momentum with the eigen value $\hbar\mathbf{k}$. The particle velocity is given by

$$v = \frac{\hbar k}{m} \tag{5.32}$$

In a system of N free electrons, the occupied states or orbitals in the ground state may be represented by points inside a sphere in the **k**-space. The energy corresponding to the surface of the sphere then represents the Fermi energy, Let $\mathbf{k_F}$ be the wave vector from the origin of the **k**-space to the surface of the sphere as shown in Fig. 5.8. Then, using Eq. (5.25), the Fermi energy is written as

$$E_F = \frac{\hbar^2}{2m} k_F^2 \tag{5.33}$$

It is observed from Eqs. (5.24) and Fig. 5.8 that there is one allowed wave vector or one distinct triplet of quantum numbers k_x, k_y and k_z which corresponds to the volume element $(2\pi/L)^3$ of **k**-space. Thus, in the sphere called the *Fermi sphere* of volume $(4\pi/3)k^3_F$, the total number of electronic states or orbitals is

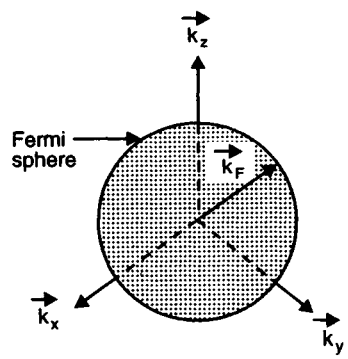


Fig. 5.8. *A sphere of radius k_F in the* **k***-space containing points which represent the occupied states of a system of N free electrons in the ground state.*

$$2\,\frac{\frac{4\pi}{3}k_F^3}{\left(\frac{2\pi}{L}\right)^3} = \frac{L^3}{3\pi^2}k_F^3 = \frac{V}{3\pi^2}k_F^3 = N \tag{5.34}$$

where the factor 2 appears because there are two allowed values of m_s, and hence two electronic states, corresponding to each value of k. The number of states has been set equal to the number of electrons, N. From Eq. (5.34), we obtain

$$k_F = \left(\frac{3\pi^2 N}{V}\right)^{1/3} \tag{5.35}$$

This shows that the value of k_F depends on the concentration of electrons, N/V, and is independent of the mass of the electron. The expression for the Fermi energy is obtained by using Eq. (5.35) in (5.33) as

$$E_F = \frac{\hbar^2}{2m}\left(\frac{3\pi^2 N}{V}\right)^{2/3} \tag{5.36}$$

i.e., E_F depends on both the electronic concentration and mass. The total number of electrons is, therefore, given by

$$N = \frac{V}{3\pi^2}\left(\frac{2mE_F}{\hbar^2}\right)^{3/2} \tag{5.37}$$

The electron velocity, v_F, at the Fermi surface is obtained from Eqs. (5.32) and (5.35) as

$$v_F = \frac{\hbar k_F}{m} = \frac{\hbar}{m}\left(\frac{3\pi^2 N}{V}\right)^{1/3} \tag{5.38}$$

The density of states function, $D(E)$, is obtained by using the fact that, in the ground state of the system, all the energy states below E_F are occupied and the total number of states is equal to the total number of electrons, i.e.,

$$\int_0^{E_F} D(E)dE = N \tag{5.39}$$

Substituting the value of N from Eq. (5.37), we get

$$\int_0^{E_F} D(E)dE = \frac{V}{3\pi^2}\left(\frac{2mE_F}{\hbar^2}\right)^{3/2}$$

Expressing the integral in an indefinite form, we obtain

$$\int D(E)dE = \frac{V}{3\pi^2}\left(\frac{2mE}{\hbar^2}\right)^{3/2} = \int \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} E^{1/2}\,dE$$

or

$$D(E) = \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} E^{1/2} \tag{5.40}$$

The variation of $D(E)$ with E is parabolic and is shown in Fig. 5.9. The value of $D(E)$ also increases with increase in crystal volume; this is to accommodate the total number of electrons N which also increases [Eq. (5.37)] with volume of the crystal. At the Fermi surface, the density of states is obtained by putting $E = E_F$ in Eq. (5.40), i.e.,

$$D(E_F) \equiv \frac{dN}{dE_F} = \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} E_F^{1/2} \tag{5.41}$$

This result may also be obtained and expressed in a simple way by taking the natural logarithm of the expression (5.37), i.e.,

$$\ln N = \frac{3}{2}\ln E_F + \text{constant}$$

or

$$\frac{dN}{N} = \frac{3}{2}\frac{dE_F}{E_F} \tag{5.42}$$

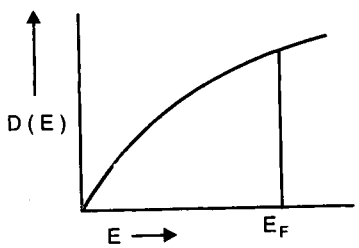


Fig. 5.9. *Variation of density of electronic states with energy for a free electron gas in three dimensions.*

Thus, within a factor of the order of unity, the number of electronic states or orbitals per unit energy range at the Fermi energy is equal to the total number of conduction electrons divided by the Fermi energy.

The density of filled electronic states, $N(E)$, at any temperature T is obtained by multplying the density of states, $D(E)$, at 0 K by the probability of occupation, $f(E)$, of the quantum state E at that temperature, i.e.,

$$N(E) = D(E)\, f(E) \tag{5.43}$$

The function $f(E)$ is the Fermi function and is defined by Eq. (5.28). The number of filled electronic states in the energy range E and $E + dE$ at temperature T is then given by

$$N(E)dE = D(E)\, f(E)dE \tag{5.44}$$

We now discuss some parameters of free electron gas at absolute zero.

(a) The Fermi Energy, E_{F_0}

The total number of electrons can be obtained by integrating Eq. (5.43), i.e.,

$$N = \int_0^\infty N(E)dE = \int_0^\infty D(E)f(E)dE \tag{5.45}$$

Using the expression (5.40), we get

$$N = \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\int_0^\infty E^{1/2} f(E)dE$$

$$= \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\left[\int_0^{E_{F0}} E^{1/2} f(E)dE + \int_{E_{F0}}^\infty E^{1/2} f(E)dE\right.$$

Now using Eqs. (5.29), we get

$$N = \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\int_0^{E_{Fo}} E^{1/2}dE$$

$$= \frac{V}{3\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} E_{Fo}^{3/2}$$

or
$$E_{Fo} = \frac{\hbar^2}{2m}\left(\frac{3\pi^2 N}{V}\right)^{2/3} \tag{5.46}$$

Putting $N/V = n$, the electron concentration, we obtain

$$E_{Fo} = \frac{\hbar^2}{2m}\left(3\pi^2 n\right)^{2/3} = \frac{h^2}{8m}\left(\frac{3n}{\pi}\right)^{2/3} \tag{5.47}$$

This shows that the Fermi energy at 0 K can be calculated simply from the electron concentration n. We also conclude from the above relationship that the electrons have some definite energy even at absolute zero. This is something entirely different from the classical theory according to which the electrons should have zero energy at absolute zero.

(b) Average Kinetic Energy, $\bar{E}_0$

The average kinetic energy of an electron at absolute zero is calculated as

$$\bar{E}_0 = \frac{1}{N}\int_0^\infty ED(E)f(E)dE$$

Using the expression (5.40) for $D(E)$, we obtain

$$\bar{E}_o = \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\frac{1}{N}\left[\int_0^{E_{Fo}} EE^{1/2}f(E)dE + \int_{E_{Fo}}^\infty EE^{1/2}f(E)dE\right]$$